08 September 2025 ASSIGNMENT by 23:59 PM

- Due Monday by 23:59
- Points 70
- Submitting on paper
- Available 6 Aug at 0:00 - 8 Sep at 23:59

COMP 1102 MATHS FOR COMPUTING

ASSIGNMENT QUESTION [TOTAL 70 MARKS]

INDIVIDUAL ASSIGNMENT – COMP1102 Maths for Computing LEARNING OUTCOMES

- Practice basic arithmetic and algebraic concepts in the aspect of computer Science.
- Apply logic that are fundamental concepts used in the functioning of computer hardware and design and programming of software.
- Apply proof techniques as computer science theory heavily depends on proof.
- Use basic abstract and linear algebra like groups and using matrices to solve systems of linear equations.
- Apply probability and concepts in Statistics.

ASSIGNMENT INSTRUCTIONS

1. You may discuss the assignment to understand the problems and the various learning topics. However, you are not allowed to share solutions to the given assignment task or your solutions with any other classmates.

1. **Citing your sources:** If you have reused any mathematical tools or techniques, you must provide the reference list in your assignment submission You do not have to cite module lecture notes, slides, module textbook, or other lab task exercises.

- **Please show your work and all the Partial credit cannot be given for a wrong answer if your work isn't shown.**

SUBMISSION REQUIREMENT

Please note that learners must submit all assignment by the stated due date (s). Your solutions to the assignment will be submitted as a single MS Word document.

File name: ***StudentNo-FullName- ISDM-assessment.docx***

1. Submit your completed report to the Canvas LMS under the folder titled "Maths for Computing" Assignment – Submission Folder" by

Submission Date: **LESSON 10:**

Submission Time: **on or before 23:59**

Late Submission maximum 3 days from the submission due date: A deduction of 5% from your actual mark shall be imposed on each day. Assignment submitted after 3 days from the due date is marked as a fail grade. No assignment will be accepted for marking after this date.

1. **Backup**
   - All assignments submitted will not be returned to learners.
   - Please ensure that backup copies are kept on different devices.
   - Learners are responsible for their own work and must perform the necessary backups.

Assignment Submission Cover Page

Student Name:	
Student ID:	
Module Name:	**COMP 1102 Maths for Computing**
Course Title	

Lecturer:	**Ms.Rathina Grace Monica Abraham Geoffrey**
Submission Date:	
Assignment Report Total number of pages	

I. **Overview**

The goal of this assignment is to allow you on how to use computer-based number systems, abstract language, work with algorithms, self-analyse your computational thinking, and accurately modelling real-world solutions.

II. Short Answer Response Questions

Attempt all the following questions and clearly type the steps showing working on the solution.

1. A register in a computer contains binary The contents of the register could represent a binary integer. Convert the binary integer to denary and hexadecimal.

1000 1111 1010 1011 (tel:1000 1111 1010 1011)

1101

2. Convert the given ASCII value to binary and hexadecimal

ASCII: Good Luck

3. The MAC address of a device is represented using Each pair of hexadecimal digits is stored using 8-bit binary.

Complete the table to show the 8-bit binary equivalents for the section of MAC address – 8B EA 9C 59

8B	**EA**	**9C**	**59**

4. Explain why hexadecimal is used to represent colour codes in HTML and CSS.
5. Convert the binary number 11101011 to decimal.
6. Convert the octal number 503 642 to decimal number.
7. Covert the binary number **11011011101101 (tel:11011011101101)** to an octal and decimal number.
8. Convert the decimal number 605 to binary.
9. Convert the hexadecimal number 8C5E to octal.
10. Convert the binary number **100011101111001 (tel:100011101111001)** to hexadecimal.
11. Convert the decimal number 5372 to hexadecimal and octal.
12. Convert the hexadecimal number 9A5F to decimal.
13. (a) Translate the following English sentences to Propositional Propositions:

(R)aining

Lynn is (S)ick

Lynn is (H)ungry

Lynn is (HA)appy

Lynnowns a (C)at

Lynn owns a (D)og

(i) It is raining if and only if Lynn is sick

(ii) If Lynn is sick then it is raining, and vice versa

(iii) It is raining is equivalent to Lynn is sick

(iv) Lynn is hungry but happy

(v) Lynn either owns a cat or a dog

(b) Translate the following Propositional Logic to English sentences. (6 marks) Let:

E=John is eating

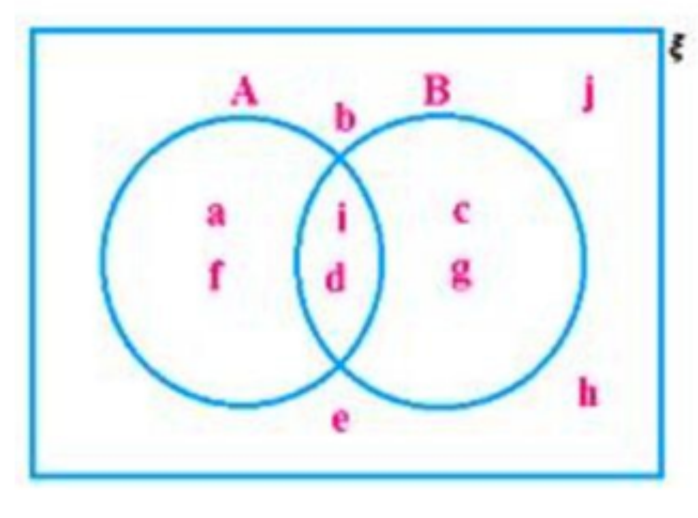
H=John is hungry

(i) E → ¬H

(ii) E ^ ¬H

(iii) ¬(H → ¬E)

14. Read the following Venn diagram and answer the following:



(i) B’

(ii) AUB

(iii) A∩ B

(iv) (AUB)’

(v) (A∩ B)’

(vi) A-B

(vii) B-A

15. Draw the Venn diagrams that represent the following

(i) A ∩ B ∩ C’

(ii) A ∩ B’∩ C’

(iii) A U B ∩ C’

16. Construct truth tables for the following propositions:

(a) Prove that *p*Ú*q* Û Ø(Ø*p* Ù Ø*q*)

(b) ~$[(p \wedge q) \rightarrow (r \wedge q)]$

(c) Show that $[(p \vee q) \wedge \neg p] \rightarrow q$ is a tautology.

17. Multiplying Polynomials:

(i) (2a − 8b)(6a − 8b)

(ii) (7m − 5n)(2m + 5n)

18. (a) What is the probability of getting a sum of 8 with a toss of two dice?

(b) Suppose we toss a coin and then a What is the sample space for this experiment? What is the probability of tossing a ((T, 3))?

(c) What is the probability that a card drawn from a pack of 52 cards will be a diamond or a king?

(d) A bag contains 6 red balls, 8 yellow balls and 4 green balls. One ball is taken random from the bag.

What is the probability that the ball is:

(i) yellow?

(ii) green?

(iii) yellow or green?

19. Find the mean and standard deviation of the weights of the 100 Male students at LSBF school as shown below .

Mass(kg)	Number of students
60 - 62	5
63 - 65	18
66 - 68	42
69 - 71	27

72 - 74	8

III. Deliverable

1. Provide at least **two** references for **each** of the following academic sources using Harvard Referencing guidelines to support your analysis:
   1. Books from the National Library ebook/e-resources
   2. E-books from DOAB - Directory of Open Access Books **https://www.doabooks.org/ (https://www.doabooks.org/)**
   3. Journal articles from DOAJ- Directory of Open Access Journals **https://doaj.org/ (https://doaj.org/)**

Refer to Harvard Referencing guideline: **https://www.citethisforme.com/harvard- (https://www.citethisforme.com/harvard-referencing) referencing (https://www.citethisforme.com/harvard-referencing)**

1. Assignment document with clearly illustrated steps for Clear and coherent description for writing your reasons.

- The cover page, table of contents, list of references, and appendices are NOT included in the word count.
- You must show your work and all the Partial credit cannot be given for a wrong answer if your work isn't shown.

IV. Marking Guide

Questions	Description	Marks Allocated	Marks Awarded
1 -12	Number and number systems *(1 marks *14; show all working steps)*	14	
13-14	Propositional and Predicate Logic	16	

	*(2 marks *8; show all working steps)*		
15	Set Theory and Combinatorics *(1 marks *10; show all working steps)*	**10**	
16	Proof *(2 marks *3; show all working steps)*	**6**	
17	Abstract and Linear Algebra *(1 marks *2; show all working steps)*	**2**	
18	Probability *(3 marks *12; show all working steps)*	**12**	
19	Statistics (10 marks) *(10 marks; show all working steps)*	**10**	
TOTAL		**70**	
OVERALL GENERAL FEEDBACK			